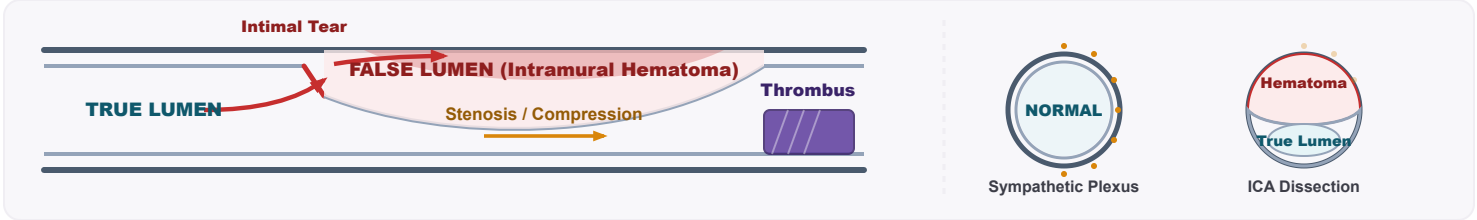


Cervical Artery Dissection

Clinical presentation, diagnostic workup, medical management (extracranial vs. intracranial), and landmark trial evidence.



1. Clinical Presentation & Pathophysiology

Ipsilateral Headache & Pain

- **Carotid (ICA):** Frontotemporal, retro-orbital, or facial pain (jaw angle).
- **Vertebral (VA):** Severe occipital headache or posterior neck pain.
- **Onset:** Pain precedes TIA/stroke by hours to days (median 4 days).

Anhidrosis-Sparing Horner's

- **Signs:** Ptosis & miosis (~28–58% of ICA) **without** anhidrosis.
- **Mechanism:** Sudomotor (sweat) fibers ascend via the ECA plexus; pupillary/eyelid fibers ascend via the ICA.

Neurological Deficits

- **CN Palsies:** CN IX, X, XI, XII palsies (~8–16%) from local ICA compression.
- **VA Territory:** Wallenberg syndrome, cerebellar ataxia, PICA/AICA territory strokes.

2. Diagnostic Workup

- **CTA Head/Neck:** First-line. Shows tapered stenosis ("string sign"), flap, pseudoaneurysm, or occlusion.
- **MRI Neck (T1 Fat-Sat):** Pathognomonic hyperintense crescentic intramural hematoma (crescent sign).
- **DSA:** Gold standard; reserve for diagnostic doubt or stenting evaluation.
- **Screening:** Screen for FMD or connective tissue disorders if spontaneous/recurrent.

3. Medical Management: Extracranial vs. Intracranial Dissection

Extracranial Dissection

- **Antithrombotics:** ≥ 3 months (Class I).
- **Aspirin vs. Anticoagulation:** Equipose. Choice between monotherapy/DAPT vs. VKA/DOAC is individualized.
- **STOP-CAD Strategy:** In occlusions, consider anticoagulation for Day 1–30, then switch to antiplatelet.
- **IV Thrombolysis:** Safe & indicated within 4.5 hours (Class I).

Intracranial & Pseudoaneurysms

- **SAH Risk:** Intracranial vessels lack external elastic lamina & have thin adventitia; high rupture/SAH risk.
- **Anticoagulation:** Contraindicated if SAH present. Screen with CT/LP first. Prefer single antiplatelet if no SAH.
- **Pseudoaneurysm:** Managed conservatively with serial imaging.
- **Stenting:** Reserve for refractory ischemia or enlarging/symptomatic pseudoaneurysms.

4. Landmark Trial & Cohort Evidence

Study / Year	Population & Design	Interventions Compared	Key Outcomes & Clinical Nuance
CADISS 2015	N = 250. Extracranial CeAD. RCT.	Antiplatelet vs. Anticoagulant for 3 months.	• Composite (Stroke/Death at 3m): 2.0% vs. 1.0% (\$p=0.63\$). No significant difference; established clinical equipoise.
TREAT-CAD 2021	N = 194 (PP = 173). Extracranial. Aspirin 300mg daily vs. VKA for 3 months. RCT.	Aspirin 300mg daily vs. VKA for 3 months.	• Composite (Stroke, bleed, death, or MRI at 14d): 23% vs. 15% (Non-inferiority NOT met). Ischemic stroke alone: 8.0% vs. 0%.
Kaufmann IPD 2024	N = 444. Meta-analysis of CADISS + TREAT-CAD.	Antiplatelet vs. Anticoagulant.	• Ischemic Stroke alone: Significant reduction with anticoagulation (0.5% vs. 4.0%; OR 0.14, \$p=0.01\$). No difference in composite endpoints.
STOP-CAD 2024	N = 3,636. Multicenter cohort registry.	Antiplatelets vs. Anticoagulation.	• Stroke vs Bleed: Anticoagulation associated with lower stroke rate but higher bleed. Occlusive cases benefited most. Proposes Day 30 transition.

CADISS: *Lancet Neurol.* 2015;14(4):361-7. PMID: 25684164 | **TREAT-CAD:** *Lancet Neurol.* 2021;20(5):341-350. PMID: 33765420
Kaufmann IPD: *JAMA Neurol.* 2024;81(6):630-637. PMID: 38739383 | **STOP-CAD:** *Stroke.* 2024;55(4):908-918. PMID: 38334460 | **AHA/ASA:** *Stroke.* 2021;52:e364-e467. PMID: 34024117